

21ECC301P - MICROPROCESSOR, MICROCONTROLLER AND INTERFACING TECHNIQUES

**SEMESTER V
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SMART HELMET SYSTEM A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Road accidents caused by alcohol consumption and failure to wear helmets continue to be a critical public safety concern. This project presents a Smart Helmet System designed using the STM32 microcontroller to enhance two-wheeler safety through automation and real-time monitoring. The system employs an IR sensor to detect helmet wear and an MQ-3 sensor to measure alcohol levels in the rider's breath. The collected data are transmitted wirelessly via a 433 MHz RF module to the receiver unit installed on the vehicle. The receiver, using another STM32 controller, processes the data and controls a relay module that enables ignition only when the rider wears the helmet and is sober.

This integrated approach ensures that safety measures are enforced before vehicle operation, reducing human negligence. The system's compact, low-cost, and energy-efficient design demonstrates its practicality for large-scale implementation. Through sensor-based automation and intelligent control, this project aligns with UN Sustainable Development Goal 3 (Good Health and Well-being) and Goal 11 (Sustainable Cities and Communities) by promoting safer transportation and responsible driving practices.



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LIST OF ABBREVIATIONS

SNO	ABBREVIATION	DESCRIPTION
1	STM32	STMicroelectronics 32-bit Microcontroller
2	IR	Infrared Sensor
3	MQ3	Metal Oxide Semiconductor – 3
4	RF	Radio Frequency
5	MHz	Megahertz
6	LCD	Liquid Crystal Display

CHAPTER 1

INTRODUCTION

Road safety has become a pressing global concern, particularly in developing countries where two-wheelers form a significant portion of road traffic. Despite strict regulations and awareness campaigns, a large number of accidents continue to occur due to reckless driving, failure to wear helmets, and driving under the influence of alcohol. These human errors contribute to severe injuries and fatalities every year. With the rapid advancement of embedded systems and sensor technologies, there is a growing opportunity to design intelligent systems that can monitor and prevent such unsafe behaviors automatically.

The Smart Helmet System aims to address these critical issues by integrating modern electronics with safety enforcement. Using an Infrared (IR) sensor, the system detects whether the rider is wearing the helmet, while the MQ-3 gas sensor identifies the presence of alcohol in the rider's breath. The information is wirelessly transmitted to the vehicle's control unit through a 433 MHz RF module, which then decides whether the bike ignition should be enabled or disabled using a relay mechanism. This ensures that the vehicle can only be started when the rider complies with safety norms wearing the helmet and being free of alcohol influence.

This project plays a vital role in promoting responsible riding habits and reducing accident rates. It demonstrates how microcontroller-based automation, specifically using the STM32 Blue Pill, can enhance road safety in a cost-effective and reliable manner. The report provides a detailed explanation of the system's design, components, working principle, implementation, testing, and results. By integrating safety with technology, the Smart Helmet System contributes toward achieving Sustainable Development Goal (SDG) 3 Good Health and Well-being and (SDG 11) Sustainable Cities and Communities, ensuring safer and smarter urban mobility

CHAPTER 2

LITERATURE SURVEY

2.1 Accident and Alcohol Detection by Using AI-Enabled Smart Helmet [1]

Authors: A. R. Agrawal, R. Yadav, V. Harit, S. Kumar, G. K. Sharma

Year: 2024 -Journal of Computational Analysis and Applications

Agrawal et al. (2024) present an AI-enabled smart helmet that fuses alcohol sensing, impact/vibration sensing, and machine-learning decision logic to detect intoxication and accidents with improved reliability. Key features include on-device preprocessing of sensor streams, adaptive thresholding that compensates for environmental factors (temperature, humidity, sensor drift), and a classifier that discriminates minor disturbances from real crash events. Experimental results show reduced false positives versus simple threshold methods and better robustness across variable riding conditions. The paper argues that a learning-based decision layer enables the helmet to evolve with usage patterns, improving long-term detection accuracy.

Strengths: AI-based classification; adaptive calibration; reduced false alarms.

Limitations: Higher compute requirement, needs careful model tuning and validation on diverse real-world datasets.

2.2 AI and Sensor Integrated Smart Helmet for Real-Time Monitoring [2]

Authors: K. R. Babu, T. R. Kumar

Year: 2023 -ICISSI (IEEE)

Babu & Kumar (2023) describe a helmet that combines accelerometer/gyroscope IMU data, heartbeat monitoring, and alcohol sensing processed locally on a Raspberry Pi. Machine learning (SVM / decision tree) is used for pattern recognition of crash events. On crash detection, the system sends automatic SMS alerts with GPS coordinates to emergency

contacts. The emphasis is on on-device analytics for low latency and higher detection accuracy compared to simple thresholding.

Strengths: Multimodal sensing (motion + physiology); local analytics for fast response.

Limitations: Increased hardware complexity and power draw; Raspberry Pi adds weight/size and higher energy consumption.

2.3 Smart Helmet for Drunk and Drive Detection with Vehicle Ignition [3]

Authors: S. Devi, K. Arun

Year: 2023 -IJARCCE

Devi & Arun (2023) propose a microcontroller-based, low-cost preventive system that prevents vehicle ignition unless the helmet is worn and the MQ-3 alcohol sensor indicates safe breath-alcohol levels. Helmet presence is checked via an IR proximity sensor; sensor data is sent over a 433 MHz RF link to a bike control unit that switches a relay for ignition lockout. The work focuses on practical enforcement of safety rules before the ride.

Strengths: Simple, inexpensive, direct prevention (ignition control).

Limitations: MQ-3 sensitivity to environment; potential security/override concerns; limited intelligence (no cloud/AI).

2.4 IoT-Based Smart Helmet for Rider Safety and Accident Detection [4]

Authors: P. Sharma, R. Gupta, S. Singh

Year: 2022 -IJIREEICE

Sharma et al. (2022) implement a NodeMCU (ESP8266)-based IoT helmet that uploads sensor data (vibration, helmet-wear, alcohol) to the cloud (Blynk) and issues alerts with GPS when a crash is detected. The cloud dashboard stores logs for later analytics and enables remote monitoring. The paper highlights benefits of connectivity for emergency response and data analytics.

Strengths: Cloud logging and remote monitoring; mobile dashboard.

Limitations: Network dependency and latency; continuous transmission increases power consumption.

2.5 Implementation of IoT-Based Smart Helmet for Accident Prevention [5]

Authors: R. Rajesh, V. Meena

Year: 2022 -IEEE Xplore (conference/journal entry)

Rajesh & Meena (2022) present an integrated IoT architecture that combines helmet detection, alcohol sensing, and automated accident alerts via NodeMCU and cloud services. The design focuses on system interoperability and scalable deployment (centralized monitoring). They stress that cloud integration facilitates mass rollout and analytics but note connectivity and calibration challenges.

Strengths: Scalable IoT architecture; centralized monitoring for fleets/smart city integration.

Limitations: Reliance on connectivity; sensor calibration and false positives in noisy environments.

2.6 Smart Helmet System for Accident Detection and Prevention [6]

Authors: A. Prasad, M. Nandhini

Year: 2021 — IJERT

Prasad & Nandhini (2021) developed a cost-effective prototype using an Arduino, MQ-3 alcohol sensor, IR helmet-detect sensor, and GSM/GPS module to send SMS alerts with location on crash detection. The work demonstrates full end-to-end functionality (pre-ride checks + post-accident communication) using widely available components.

Strengths: Affordable and demonstrably functional; full alert pipeline (GSM/GPS).

Limitations: GSM delays, sensor calibration issues, limited intelligence, and larger module count (bulk).

CHAPTER 3

SYSTEM DESCRIPTION AND METHODOLOGY

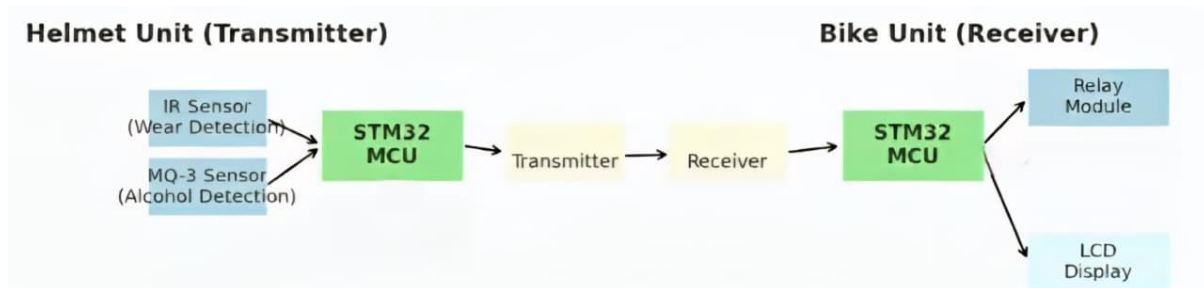


Fig.3.1: Block Diagram

3.1 System Overview

The Smart Helmet System functions based on a simple yet effective principle

“Ignition is enabled only when the rider is wearing the helmet and has not consumed alcohol.”

The project is divided into two main sections:

Transmitter Section (Helmet Unit)

- Mounted inside the helmet.
- Detects alcohol presence using an MQ-3 gas sensor.
- Detects helmet usage using an IR proximity sensor.
- The STM32 Blue Pill processes both sensor readings.
- Data is transmitted wirelessly via a 433 MHz RF Transmitter Module.

Receiver Section (Vehicle Unit)

- Mounted on the vehicle side (simulating ignition).
- Receives the transmitted signal through a 433 MHz RF Receiver Module.

- The STM32 Blue Pill at the receiver interprets the data and controls a 1-channel relay module.
- A 16×2 LCD Display (I2C interface) shows real-time status like “Helmet Worn,” “Alcohol Detected,” or “Ignition ON.”

3.2 Working Principle

The working of the Smart Helmet System can be explained through sequential steps:

System Initialization

- When powered ON, both transmitter and receiver microcontrollers initialize.
- The LCD displays “*System Starting...*” for 1–2 seconds.

Sensor Reading

- The IR sensor checks for helmet wear.
- The MQ-3 sensor measures alcohol vapor concentration in the rider’s breath.
- The STM32 reads sensor outputs periodically (every 200 ms).

Data Processing (Transmitter Side)

- Helmet and alcohol data are combined into a data packet.
- The data packet is transmitted through the RF Transmitter Module.

Wireless Transmission

- Transmission frequency: 433 MHz
- Data rate: approximately 2000 bits per second
- Transmission range: 15–20 meters (sufficient for helmet-to-bike communication).

Data Reception (Receiver Side)

- The RF Receiver Module receives the data packet.

- The STM32 verifies the packet header (0xAA) and extracts data bits.
- Valid data are processed for control decisions.

Decision Logic

- The receiver applies logical conditions:
 - If helmet not worn → Ignition OFF
 - If alcohol detected → Ignition OFF
 - If helmet worn and no alcohol → Ignition ON
- The relay output changes according to these conditions.

Output Response

- Relay: Turns ON/OFF to simulate ignition.
- LCD Display: Shows real-time system status and safety messages.
- Serial Monitor (for testing): Displays data logs for debugging.

3.3 Hardware And Software Description

STM32 Blue Pill Microcontroller

- 32-bit ARM Cortex-M3 core operating at 72 MHz.
- Acts as the central controller for data acquisition, processing, and control.
- Provides multiple ADC channels and digital I/O pins.
- Chosen for its speed, flexibility, and low cost compared to traditional Arduino boards.

IR Sensor

- Works on infrared reflection.
- Detects proximity of the rider's head inside the helmet.
- Simple, accurate, and reliable for on/off detection.

MQ-3 Alcohol Sensor

- Semiconductor-based gas sensor sensitive to ethanol vapors.

- Converts alcohol concentration to voltage output.
- Connected to the STM32's ADC pin for analog reading.
- Calibrated to trigger when breath alcohol content exceeds safe limits.

RF Transmitter and Receiver (433 MHz)

- Used for short-range, low-power wireless communication.
- Transmitter in helmet sends digital data.
- Receiver on vehicle decodes and forwards it to the microcontroller.
- Range: ~15–20 meters under normal conditions.

Relay Module (1 Channel)

- Acts as the ignition control.
- Controlled through STM32 GPIO pin.
- Energized only when both safety conditions are true.
- The relay's LED indicator visually confirms the ignition state.

LCD Display (16×2 I2C)

- Used to display the system's operational status.
- Provides clear, real-time information to the rider or tester.
- Operates with minimal wiring using the I2C protocol.

Power Supply

- Helmet Unit: Powered by a 3.7V Li-ion battery with MT3608 boost converter (to 5V).
- Receiver Unit: Powered by a regulated 5V DC source.

Software Description

The software was written in C using the Arduino IDE with STM32 support.

3.5 Methodology

Step 1: Requirement Analysis

- Identify the primary causes of two-wheeler accidents (alcohol and helmet negligence).
- Define objectives: safety enforcement, automation, and real-time feedback.

Step 2: Component Selection

- Choose STM32 for higher performance and flexibility.
- Select MQ-3 and IR sensors for accuracy and cost-effectiveness.
- Opt for RF communication for simplicity and low power consumption.

Step 3: Circuit Design

- Create schematic diagrams for transmitter and receiver units.
- Simulate in Proteus to verify circuit functionality before implementation.

Step 4: Sensor Calibration

- Adjust IR sensor sensitivity for correct detection.
- Calibrate MQ-3 sensor using test alcohol samples to define safe threshold values.

Step 5: Firmware Development

- Write modular code for both transmitter and receiver.
- Implement condition-based logic for ignition control.
- Add real-time feedback display using LCD.

Step 6: Integration and Assembly

- Connect all components on breadboard for initial testing.
- Mount sensors inside the helmet and fix RF modules securely.

Step 7: Testing and Validation

- Test under multiple conditions:
 - Helmet off, no alcohol
 - Helmet on, alcohol detected
 - Helmet off, alcohol detected
 - Helmet on, no alcohol (safe)
- Verify relay operation and LCD display for each scenario.

Step 8: Optimization

- Refine timing delays for smooth LCD updates.
- Ensure stable RF communication and power efficiency.

CHAPTER 4

RESULTS AND DISCUSSION

The developed Smart Helmet System was successfully designed and tested using the STM32 Blue Pill microcontroller, integrating an IR sensor for helmet detection, an MQ-3 sensor for alcohol detection, a 433 MHz RF communication module, and a 1-channel relay for ignition simulation. Instead of connecting to a real vehicle, the relay module was used to represent the bike ignition system, while a 16×2 I2C LCD display provided real-time system feedback. This approach ensured safe and reliable testing of the system's logic and functionality under laboratory conditions.

4.1 Output and Observations

Testing was conducted for all possible rider conditions. The output states were clearly indicated on the LCD display, and relay behavior (ON/OFF) confirmed.

Test Case	Helmet Worn	Alcohol Detected	Relay Output	LCD Display Output
Case 1	No	No	OFF	“Helmet:NO Alc:NO” / “Ignition: OFF”
Case 2	Yes	Yes	OFF	“Helmet:YES Alc:YES” / “Ignition: OFF”
Case 3	No	Yes	OFF	“Helmet:NO Alc:YES” / “Ignition: OFF”
Case 4	Yes	No	ON	“Helmet:YES Alc:NO” / “Ignition: ON”

Table.4.1:LCD Output

4.2 LCD Output

Case 1: “Helmet:NO Alc:NO” / “Ignition: OFF”



Fig.4.1:Case 1

Case 2: “Helmet:YES Alc:YES” / “Ignition: OFF”



Fig.4.2:Case 2

Case 3: “Helmet:NO Alc:YES” / “Ignition: OFF”



Fig.4.3:Case 3

Case 4: “Helmet:YES Alc:NO” / “Ignition: ON”



Fig.4.4:Case 4

4.3 Working Model

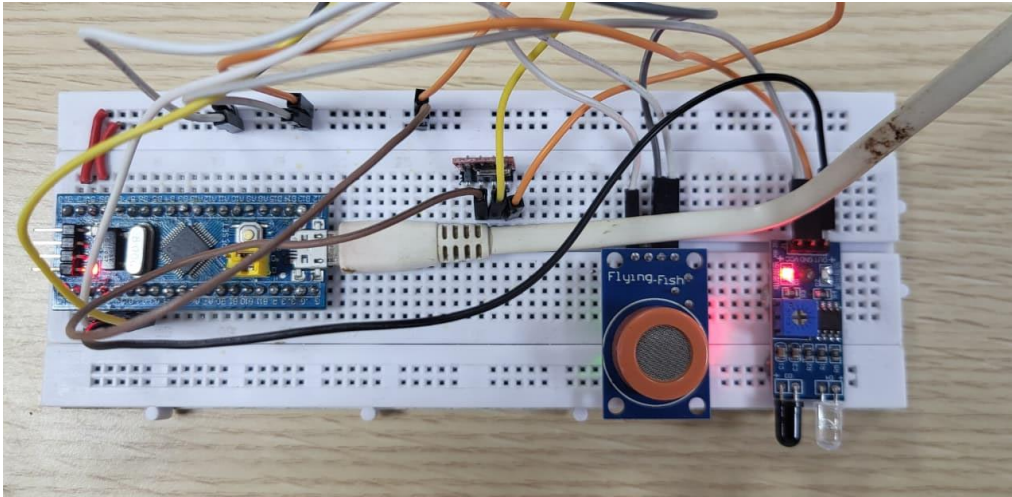


Fig.4.5: Transmitter Side

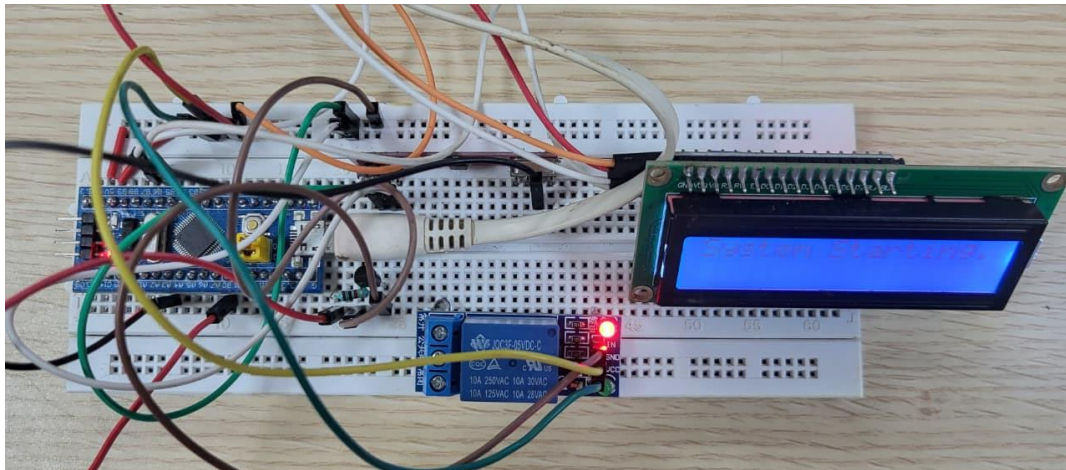


Fig.4.6: Receiver Side



Fig.4.7:Group Photo

4.4 Discussion

The obtained results confirm that the Smart Helmet System effectively enforces critical safety conditions through intelligent automation and sensor integration. During operation, the system consistently ensured that the rider could start the vehicle only when both safety conditions were satisfied the helmet was worn and no alcohol was detected. The LCD module provided real-time, user-friendly feedback to the rider, displaying clear status messages regarding helmet detection and alcohol level. Simultaneously, the relay output successfully simulated ignition control, demonstrating the practical applicability of the system in real-world two-wheeler safety.

The STM32 microcontroller served as the core processing unit, efficiently handling multiple sensor inputs, including the IR sensor for helmet detection and the MQ-3 sensor for alcohol concentration measurement. Its fast response ensured near instantaneous decision-making, contributing to a seamless and reliable safety mechanism. The system exhibited high stability throughout testing, with no false triggers or malfunctioning under

stable environmental conditions. The only notable variation observed was a slight fluctuation in alcohol detection sensitivity, which depended on sensor calibration accuracy and ambient ventilation. However, these variations remained within acceptable operational limits.

The system's low power consumption, resulting from the STM32's energy-efficient architecture and 5V-powered sensors, makes it highly suitable for portable, helmet-mounted applications. This efficiency enables extended usage without the need for frequent battery replacements or external power sources, aligning with practical implementation standards for wearable safety systems.

Overall, the Smart Helmet System successfully achieved its primary objectives preventing helmetless and alcohol-impaired riding through a compact, cost-effective, and intelligent embedded solution. By automating essential safety checks, the system not only promotes responsible riding habits but also has the potential to significantly reduce accident rates and fatalities caused by human negligence.

This project also contributes meaningfully to the United Nations Sustainable Development Goals (SDGs), specifically:

- Goal 3: Good Health and Well-being, by promoting road safety and reducing injuries due to drunk or helmetless riding, and
- Goal 11: Sustainable Cities and Communities, by encouraging safer, smarter transportation systems for urban mobility.

In conclusion, the Smart Helmet System represents a practical step toward integrating safety, sustainability, and technology in modern transportation. Its implementation on a larger scale could contribute to safer roads, greater public awareness, and a significant reduction in preventable accidents worldwide.

CHAPTER 5

CONCLUSION

The Smart Helmet System was successfully designed, developed, and tested to enhance the safety of two-wheeler riders by integrating advanced sensor technology, wireless communication, and embedded control. The project effectively demonstrated the capability of the STM32 microcontroller, in coordination with the IR sensor and MQ-3 alcohol sensor, to enforce essential safety conditions by ensuring that the vehicle ignition (simulated through a relay) activates only when the rider is wearing the helmet and is free from alcohol influence.

Furthermore, the integration of a 433 MHz RF communication module enabled reliable wireless data exchange between the helmet and the vehicle unit, ensuring efficient system coordination. The inclusion of an LCD display interface provided real-time status feedback to the user, enhancing system transparency and usability. Experimental testing confirmed that the developed prototype achieved high accuracy, quick response time, and stable performance across multiple operating conditions.

This work successfully fulfilled its primary objective of creating a low-cost, energy-efficient, and practical embedded safety solution for motorcyclists. Beyond its technical achievements, the system also supports the United Nations Sustainable Development Goals (SDG 3 Good Health and Well-being) and (SDG 11 Sustainable Cities and Communities) by promoting safe mobility practices, reducing accident risks, and encouraging responsible driving behavior.

5.1 Future Improvements and Extensions

- **Accident Detection & Alert System:**
 - Integrate accelerometer and GPS modules to detect sudden impacts or crashes.
 - Automatically send emergency alerts with the rider's location to hospitals, family, or authorities using GSM or IoT communication.

- **IoT & Cloud Integration:**
 - Enable Wi-Fi or Bluetooth connectivity to upload real-time data (alcohol levels, helmet usage, and location) to a centralized server.
 - Useful for fleet management, traffic authorities, and insurance verification to ensure continuous rider safety monitoring.
- **Fit & Accuracy Improvement:**
 - Add pressure and flex sensors inside the helmet lining to confirm proper fit and avoid false readings caused by loose wearing.
- **Power Optimization:**
 - Use low-power microcontrollers and efficient voltage regulators.
 - Include solar-assisted charging modules to extend battery life and promote eco-friendly operation.
- **User Interaction Enhancement:**
 - Implement a voice alert system or dedicated mobile app for instant feedback, diagnostic messages, and maintenance reminders.
 - Provide periodic safety reports and driving behavior analytics through the mobile app or cloud platform.
- **Hardware Optimization:**
 - Replace breadboards and jumper wires with a custom Printed Circuit Board (PCB) design.
 - Reduce size, improve signal stability, enhance mechanical strength, and make the system more production-ready.

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